Bioinstrumentation Amplifier Characteristics Lab Report

Lab 1

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BME 473: BIOINSTRUMENTATION

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1. Abstract:

The purpose of this experiment was to investigate the operating characteristics of the ETH-256 bioinstrumentation amplifier using a National Instruments CompactDAQ data acquisition system and LabVIEW. Several amplifier characteristics were examined, including gain accuracy, saturation limits, filter response, and phase shift. A function generator was used to provide known input signals, while LabVIEW was used to acquire and analyze both input and output waveforms. Experimental observations demonstrated that amplifier gain closely followed the selected gain settings, although limitations such as saturation and frequency-dependent attenuation were observed. Low-pass filter performance was evaluated by measuring output amplitude and phase changes at different frequencies. These results demonstrated the practical behavior of instrumentation amplifiers used in biomedical signal acquisition systems.

2. Introduction:

Biomedical instrumentation systems are designed to acquire and process physiological signals generated by the human body. Common biological signals such as electrocardiograms (ECG), electromyograms (EMG), and electroencephalograms (EEG) typically have amplitudes in the microvolt-to-millivolt range and are often contaminated by electrical noise. As a result, amplification and signal conditioning are required before these signals can be accurately recorded and analyzed.

Differential amplifiers play an important role in Biomedical Signal acquisition, because they amplify the difference between two input signals while rejecting voltages that are common

to both inputs. This characteristic, known as common-mode rejection, enables the amplifier to reduce unwanted electrical interference while preserving the desired physiological signal. The effectiveness of this noise rejection is described by the common-mode rejection ratio (CMRR), which is commonly expressed in decibels (dB). High CMRR values are particularly important in biomedical applications because biological signals often have much smaller amplitudes than surrounding environmental noise.

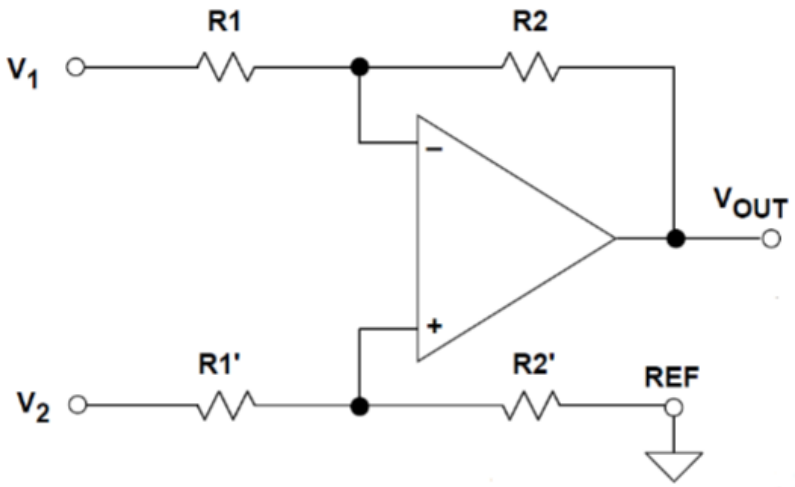


Figure 1: Differential Amplifier

Although differential amplifiers provide significant noise rejection capabilities, their performance depends heavily on precise component matching and balanced input impedances. Variations in these characteristics can reduce common-mode rejection and degrade measurement accuracy. To overcome these limitations, instrumentation amplifiers are commonly used in biomedical measurement systems.

Instrumentation amplifiers play a critical role in biomedical signal acquisition because they provide high gain, high input impedance, and improved rejection of unwanted noise. By amplifying low-amplitude biological signals while minimizing distortion, instrumentation amplifiers enable accurate measurement of physiological activity. In addition to amplification,

many biomedical amplifiers incorporate filtering circuits that limit unwanted frequency components and improve overall signal quality.

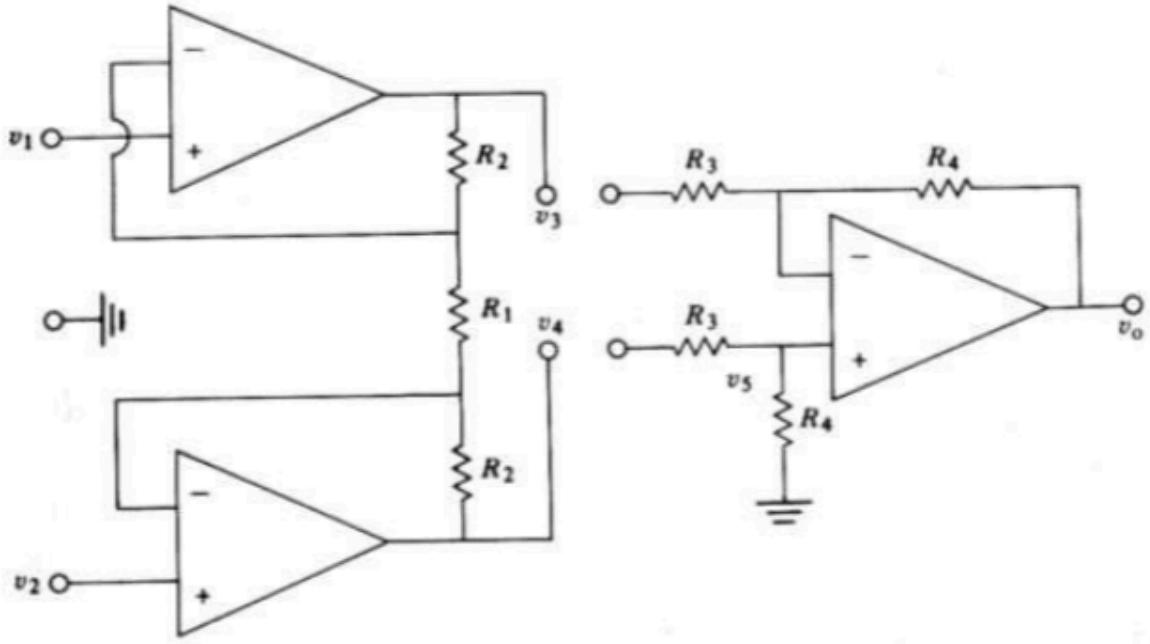


Figure 2: Instrumentation Amplifier

An important characteristic of an amplifier is its gain, which determines the relationship between the input and output signal amplitudes. Although amplifiers are often labeled with specific gain values, their actual performance must be experimentally verified. Amplifiers are also limited by saturation, which occurs when the output voltage reaches its maximum operating range and waveform distortion or clipping results. Understanding the operating limits of an amplifier is essential for obtaining accurate measurements.

Filters are another important component of biomedical instrumentation systems. High-pass and low-pass filters can be used to remove unwanted frequency components from a signal. The frequency-domain behavior of these filters can be characterized through measurements of signal attenuation and phase shift at various frequencies. These characteristics are commonly represented using Bode plots, which illustrate the relationship between frequency, gain, and phase.

The purpose of this laboratory experiment was to investigate the operating characteristics of the ETH-256 bioinstrumentation amplifier. Measurements were performed to verify amplifier gain settings, determine saturation limits, evaluate low-pass filter performance, investigate phase response, examine the response to square-wave inputs, and characterize amplifier noise. The results of these experiments provide insight into the practical behavior and limitations of instrumentation amplifiers used in biomedical signal acquisition systems.

3. Methodology:

3.1 Experimental Setup

This experiment was conducted using an ETH-256 bioinstrumentation amplifier, a National Instruments CompactDAQ (cDAQ-9174) data acquisition system equipped with an NI-9215 analog input module, a function generator, a digital oscilloscope, LabVIEW software, and BNC cables.



Figure 3: NI CompactDAQ system, including a USB chassis cDAQ-9174 (left), and an analog input module NI 9215 (right)

The NI CompactDAQ system was used to acquire and digitize the input and output signals for analysis in LabVIEW.



Figure 4: The ETH-256 Front Panel (top) and Rear Panel (bottom)

The ETH-256 amplifier was used to amplify and filter the input signals before they were acquired by the CompactDAQ system.

The function generator was initially connected to the oscilloscope using a BNC cable to verify the generated waveform. A 20 Hz sinusoidal signal with an amplitude of 50 mV peak-to-peak was selected to simulate a biological signal. After verification, the output of the

function generator was connected to both the input of the ETH-256 amplifier and the NI-9215 analog input module. The output of the amplifier was connected to a second NI-9215 input channel, allowing the original and amplified signals to be acquired simultaneously.

3.2 LabVIEW Data Acquisition Configuration

The data acquisition system was configured in LabVIEW using the NI-DAQ Assistant Express VI. An analog input channel was created to communicate with the NI-9215 module and acquire voltage measurements from the connected devices. The acquisition mode was configured for Continuous Samples operation so that signals could be monitored and recorded continuously throughout each experiment.

The two analog input channels were configured to acquire the function-generator signal and amplifier output simultaneously using the connections described in Section 3.1. This allowed the original signal to be measured directly while also being applied to the amplifier. The output of the ETH-256 amplifier was connected to a second NI-9215 input channel. As a result, both the input and amplified output signals could be acquired and analyzed simultaneously.

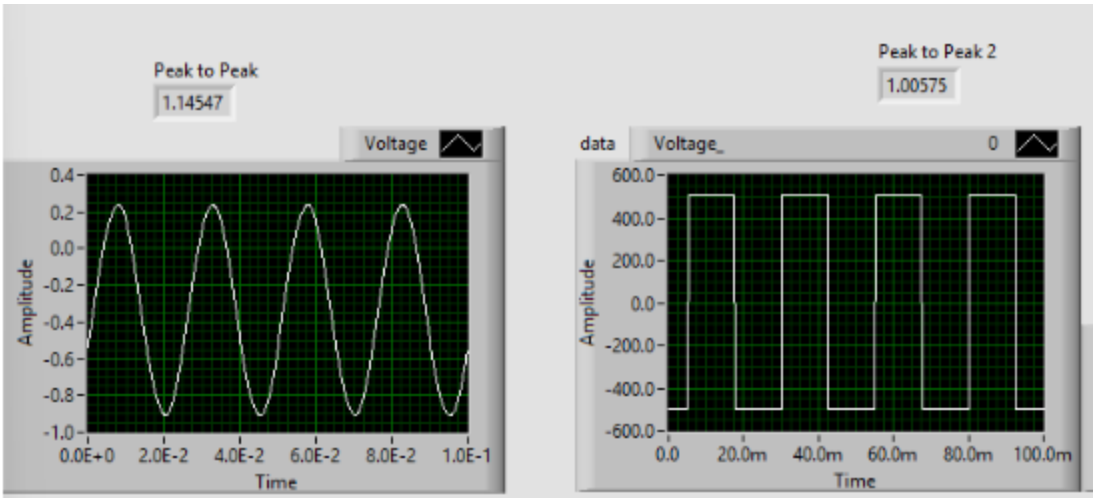


Figure 5. LabVIEW Front Panel used for real-time data acquisition and waveform visualization.

The LabVIEW front panel was then configured to display waveform graphs for both channels. Separate waveform displays allowed the input and output signals to be viewed independently even when substantial differences in signal amplitude existed. Real-time waveform monitoring was used throughout the experiment to verify proper operation of the amplifier and data acquisition system.

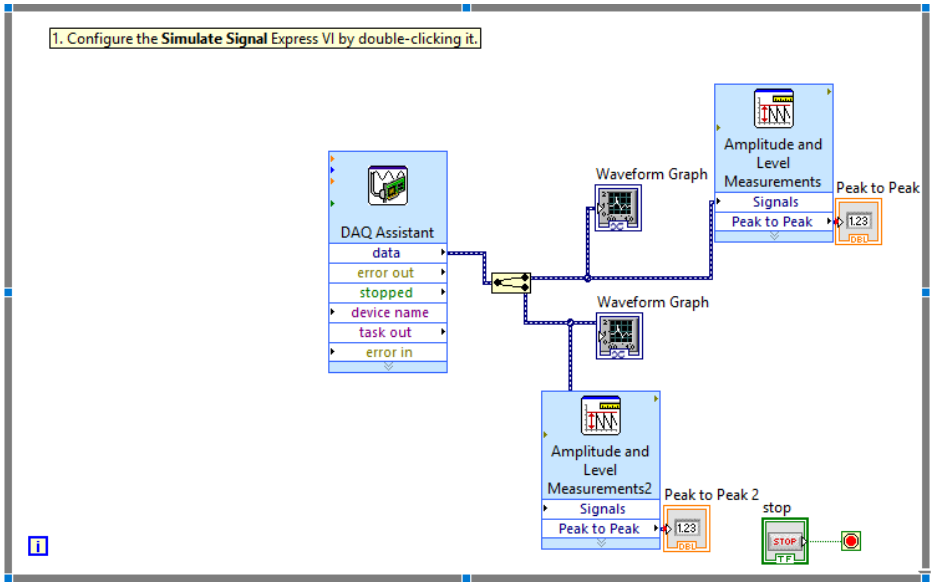


Figure 6. LabVIEW Block Diagram showing the DAQ Assistant and Split Signals Express VI configuration.

Because multiple channels were acquired through a single DAQ Assistant block, a Split Signals Express VI was incorporated into the block diagram to separate the channels into individual data streams. This enabled the input and output waveforms to be displayed on separate graphs for easier analysis. Figure 4 shows the basic acquisition configuration used to separate the two recorded channels, display their waveforms, and measure their peak-to-peak voltages. The configuration used for phase measurements is described in Section 3.6.

For the documented 10 Hz sinusoidal signal, the sampling rate was 1,000 samples every second. The signal period was .01 s, giving 100 samples/cycle. Five complete cycles took up 0.5 s and therefore corresponded to 500 samples. These relationships were used to distinguish the sampling rate, expressed in samples/second, from the number of samples acquired during each reading

At 10 Hz, the 1,000 samples/s sampling rate provided 100 samples per cycle and exceeded the Nyquist minimum of 20 samples/s. The additional samples provided a more detailed representation of the sinusoidal waveform for amplitude and phase measurements. For higher input frequencies, the sampling rate needed to be increased if the same number of samples per cycle was to be maintained.

Throughout the experiment, oscilloscope measurements were periodically compared with the LabVIEW displays to verify signal integrity and confirm proper acquisition. All acquisition parameters, including sampling rates, samples per read, gain settings, and data file names, were documented for each experiment.

3.3 Amplifier Gain Characterization

The gain settings of the ETH-256 amplifier were verified by comparing the amplitudes of the input and output signals. The amplifier high-pass filter (HPF) was set to DC, and the low-pass filter (LPF) was set to 10 kHz to provide the widest available passband and minimize filtering effects during gain measurements. Prior to data collection, the offset control on the amplifier was adjusted to remove any DC offset present in the output waveform.

A sinusoidal signal generated by the function generator was applied to the amplifier input. The signal was simultaneously acquired at the amplifier input and output through separate channels of the NI-9215 data acquisition module. Measurements were performed for amplifier gain settings of ×1, ×10, ×100, and ×1000.

For higher gain settings, particularly ×1000, the amplitude of the function generator output was reduced as necessary to prevent amplifier saturation and waveform clipping. The input and output waveforms were monitored continuously using LabVIEW and verified using the oscilloscope when required.

Peak-to-peak voltages were measured for both the input and output signals using LabVIEW measurement tools. A peak-to-peak measurement VI was used to automatically determine signal amplitudes. When necessary, waveform cursors were added to the LabVIEW graphs to manually determine peak-to-peak values. Separate cursors were positioned at the maximum and minimum points of the waveform, and the difference between these values was recorded. For low-amplitude signals, the graph display was scaled to make small voltage changes easier to read using the cursors. This adjustment improved the readability of the displayed measurements but did not change the resolution of the data acquisition hardware. Displayed values were converted to the original voltage units before calculating gain.

The measured voltage gain was calculated by dividing the output peak-to-peak voltage by the input peak-to-peak voltage

$$\text{Gain} = \frac{V_{out,p2p}}{V_{in,p2p}}$$

Both voltages were expressed in the same units before calculating the ratio. The percentage gain error was calculated relative to the selected amplifier gain

3.4 Determining Amplifier Saturation Limit

The saturation limits of the ETH-256 amplifier were evaluated for each available gain setting. The high-pass filter was set to DC and the low-pass filter was set to 10 kHz to minimize filtering effects. A sinusoidal input signal was applied to the amplifier input, and the gain control was adjusted sequentially to ×1, ×10, ×100, and ×1000.

For each gain setting, the amplitude of the function generator output was gradually increased while monitoring the amplifier output waveform in LabVIEW and on the oscilloscope. Prior to each measurement, the amplifier offset control was adjusted to minimize DC offset and center the waveform about 0 V. Input amplitude was increased until distortion of the output waveform was observed. Saturation was identified by clipping of the positive and negative peaks of the waveform. The maximum input amplitude that produced an undistorted output signal was recorded for each gain setting. Waveforms demonstrating clipping behavior were saved for subsequent analysis.

3.5 Amplifier filter testing

The frequency response of the amplifier's low-pass filter was evaluated using a sinusoidal input signal generated by the function generator. The amplifier high-pass filter was maintained at the DC setting throughout this procedure. To improve measurement accuracy and reduce the influence of noise, the input signal amplitude was increased to approximately 1 V peak-to-peak.

The low-pass filter was sequentially adjusted to cutoff settings of 5 Hz, 50 Hz, 150 Hz, 2 kHz, and 10 kHz. For each setting, the frequency of the input signal was varied while monitoring the output waveform. Output amplitude measurements were recorded at multiple frequencies. At each test frequency, the measured gain was calculated as the ratio of output peak-to-peak voltage to input peak-to-peak voltage. The cutoff frequency was identified as the frequency at which this gain decreased to approximately 0.707 times the measured passband gain, corresponding to a decrease of approximately 3.01 dB:

$$\frac{G(f_c)}{G_{\text{passband}}} = \frac{1}{\sqrt{2}} \approx 0.707$$

3.6 Bode Plot and Phase Response Measurements

Additional frequency-response measurements were performed for the 50 Hz low-pass filter configuration. The frequency of the sinusoidal input signal was varied over a range of values both below and above the filter cutoff frequency. At each frequency, the output amplitude was measured and recorded using LabVIEW.

The Tone Measurements Express VI was used to determine the phase angle of both the input and output waveforms. A Subtract VI was used to calculate the phase difference between the two signals. Amplitude and phase measurements were recorded over the selected frequency range and subsequently used to generate magnitude and phase Bode plots describing the frequency response of the filter.

3.7 Square-Wave Response Analysis

The transient response of the amplifier was investigated using a 40 Hz square-wave input signal. The square wave generated by the function generator was applied to the amplifier input while output waveforms were recorded in LabVIEW.

Measurements were first performed with the low-pass filter set to 50 Hz. The low-pass filter setting was then changed to 10 kHz, and the measurement procedure was repeated. Output waveforms from both filter settings were saved for comparison. Differences in waveform shape, rise time, and edge sharpness were documented for later analysis.

3.8 No signal noise measurements

Noise performance measurements were conducted by disconnecting the function generator from the amplifier input and connecting a 50-ohm termination to the amplifier input terminal. This configuration ensured that no intentional signal was applied to the amplifier, allowing only internally generated and environmental noise to be measured.

The resulting noise waveform was observed using both LabVIEW and the oscilloscope. Measurements were performed for multiple combinations of gain, high-pass filter, and low-pass filter settings. The dominant frequency component of the noise was identified from the acquired signals, and the approximate voltage range of the noise was measured. Waveforms were recorded for each configuration to evaluate the influence of amplifier gain and filter settings on overall noise performance.

4. Results :

Data was acquired using NI CompactDAQ and Labview. A sampling rate of 2kHz with a sample size of 200 was selected. This provided a 0.1 second acquisition window, yielding a 10Hz frequency. This rate sufficiently oversampled the 20-150 Hz test frequencies.

The amplifiers nominal gain settings were verified at 20 Hz input frequency with a 10 kHz LPF. As shown in Table 1, the measured gains closely match the panel markings:

Nominal Gain	Output Peak to Peak (V)	Measured Gain	% Error
x1	0.0535	1.00	0.0%
x10	0.4965	9.28	-7.2%
x100	4.9555	92.63	-7.4%
x1000	49.655	928.13	-7.2%

Table 1: Measured Gain vs Nominal Gain (LPF=10kHz), Input Signal: 20Hz sine wave

Sample calculation: $G_{measured} = V_{out}/V_{in} = 4.955\,V/0.0535\,V = 92.63$

When the LPF was changed to 5 Hz for the 20 Hz signal, the output amplitude was greatly reduced.

Nominal Gain	Output Peak to Peak (V)
x1	0.00601

x10	0.00443
x100	0.01171
x1000	0.04811

Table 2: Effect of LPF=5Hz on Gain, Input Signal: 20Hz sine wave

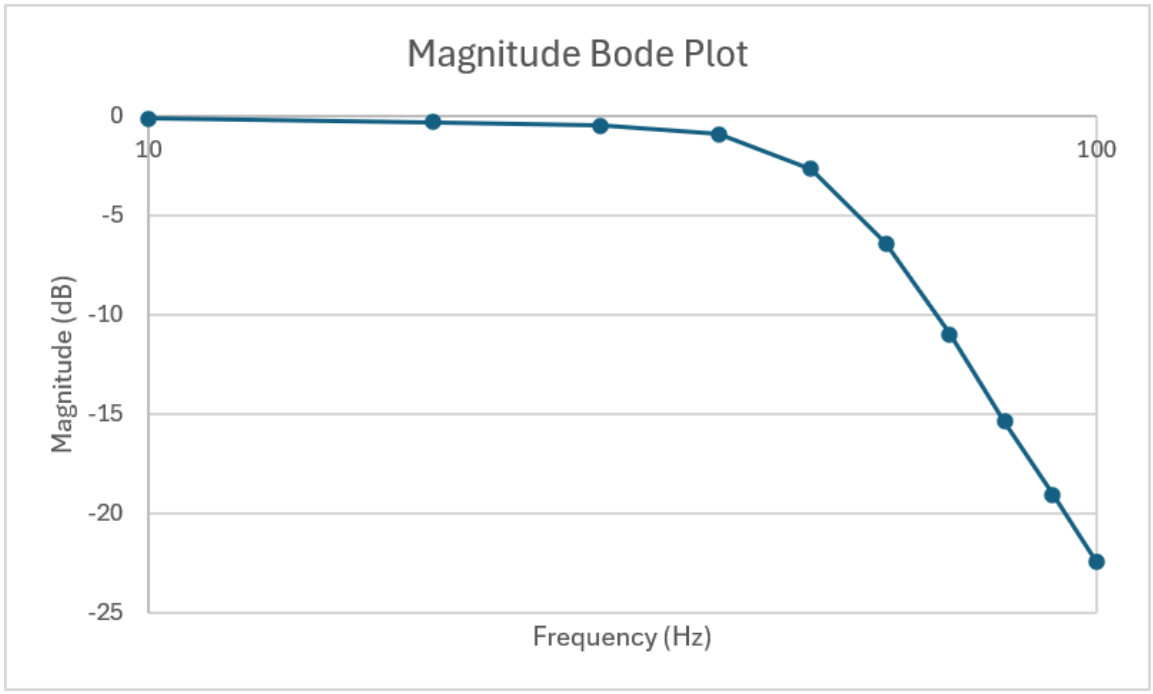
At a gain of x1000, the amplifier saturated when the input signal exceeded 20mV. Given the amplifiers approximate ±10V output rails, an input of 20mV results in the maximum linear output. Exceeding 20mV causes the signal to clip.

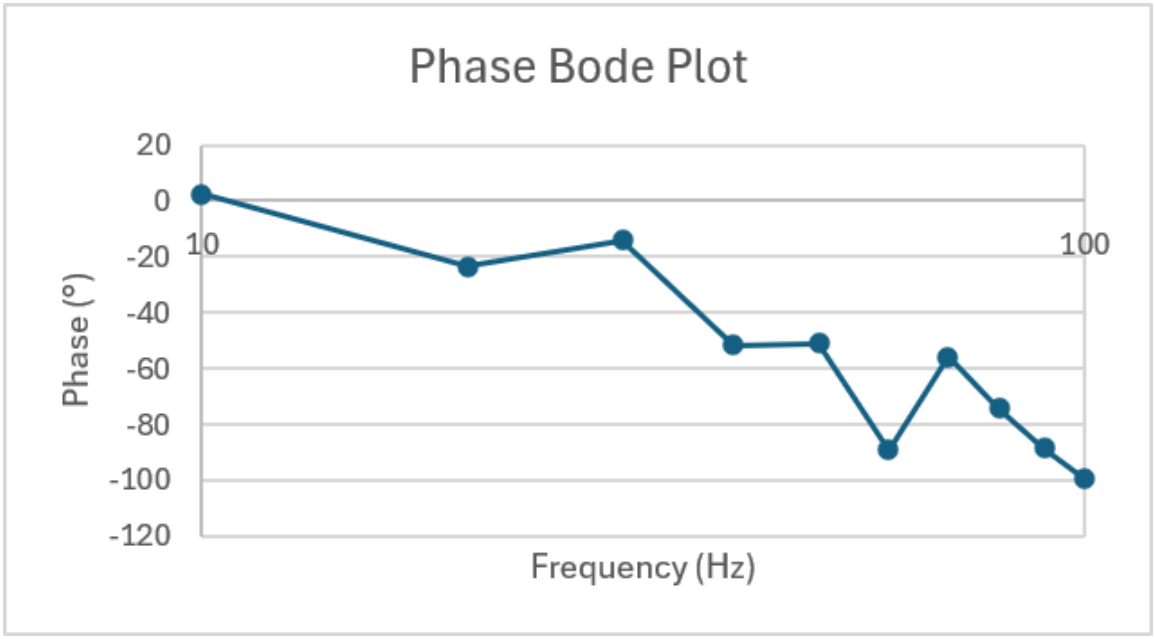
The LPF cutoff frequencies were measured and compared to nominal values

Nominal LPF Setting	**Measured Cutoff Frequency**
5 Hz	4.05 Hz
50 Hz	50.6 Hz
150 Hz	163 Hz
2 kHz	2030 Hz
10 kHz	9770 Hz

Table 3: LPF Cutoff Frequencies where cutoff was defined at 0.707 ptp output

For the 50 Hz LPF setting, a frequency sweep from 10 Hz to 100 Hz was conducted to generate Bode plots. The magnitude response and the phase response demonstrate the expected behavior of a first-order filter





Graphs: Magnitude Bode Plot(Top); Phase Bode Plot(Bottom)

A 40 Hz square wave passed through the amplifier. When the LPF was set to 50 Hz, the output amplitude was 1.1455 V with a phase angle of 51.6°, and the waveform appeared heavily distorted. When the LPF was set to 10kHz, the output amplitude was 1.0 V with a phase angle of 179.5°. The output retained the structure of a square wave.

LPF	Output PTP (V)	Phase Angle	Output Description
50 Hz	1.1455	51.633	Sine wave output
10kHz	1.00	179.544	Square wave output

Table 4: Output Characteristics of a 40 Hz square wave under two different LPF settings

With the input terminated by a 50 ohm resistor, the baseline noise was characterized. At a gain of x5000, the dominant noise component was measured at 60.98 Hz with a peak-to-peak magnitude of 206 mV

Parameter	Measured Value
Main Frequency Component	60.98 Hz
Peak to Peak Noise (Gain=5k)	206 mV

Table 5: No Signal Noise Measurements

5. Discussion :

The results showed how the amplifiers gain and filter setting effected the recorded signals. A sampling rate of 2000 samples/s and 200 samples/ read gave a recording interval of 0.1 s. For the 20 Hz input, this captured two complete cycles with 100 samples/cycle. This provided enough detail to view the sinusoidal waveform, although higher sampling rate would be needed for the 2kHz and 10kHz cutoffs.

With the LPF set to 10 kHz, the measured gains were 1.00, 9.28, 92.63, and 928.13 for the nominal ×1, ×10, ×100, and ×1000 settings. The ×10 and ×100 gains were about 7% below their expected values. Their similar percentage errors suggest that a consistent measurement issue or amplifier gain error may have contributed. However, the cause cannot be determined without checking the input voltages and repeating the measurements. The ×1000 result also needs to be checked because the recorded output was 49.655 V peak-to-peak. A scaling or unit error could affect both this value and the calculated gain.

Clipping was observed at x500, showing that increasing gain can distort the output when the amplifier reaches its voltage limit. However the input amplitude and output clipping voltages were very noisy and did not increase consistently with the gain. They show that attenuation occurred, but they are not reliable enough to calculate its exact amount.

Most measured cutoff frequencies were close enough for the selected settings. The nominal 50 Hz, 2 kHz, and 10 kHz settings produced cutoffs of 50.6, 2030, and 9770 Hz, corresponding to differences of +1.2%, +1.5%, and −2.3%. Larger differences occurred at the 5 Hz and 150 Hz settings, where the measured cutoffs were 4.05 and 163 Hz. Noise and difficulty locating the exact cutoff may have contributed to these differences. These comparisons depend

on identifying a cutoff at 70.7% of the passband output while keeping the input amplitude constant.

The 50 Hz frequency sweep showed a clear decrease in output as frequency increased. The output fell from 0.9831 V Peak to Peak at10 Hz to 0.7348 V at 50 Hz and 0.0756 V at 100 Hz. Assuming the input remained constant, this means the 100 Hz output was only about 7.7% of the 10 Hz output. The phase readings were less consistent, with large changes between some neighboring frequencies. These values need to be worked on more before drawing conclusions from them.

The squarewave test made the effect of filtering easier to see. At the 50 Hz LPF setting, the 40 Hz square wave input produced a sine-like output. At 10 kHz, the output kept it square like shape. A square wave is made up of a fundamental frequency and at higher off harmonics. For a 40 Hz signal, these harmonics include 120, 200, and 280 Hz. The 50 Hz filter attenuated these higher-frequency components, rounding the waveform. The 10 kHz setting allowed more harmonics through and preserved the sharper edges. The outputs were 1.1455 V peak-to-peak at 50 Hz and 1.00 V at 10 kHz, but this difference alone does not prove a gain error because the waveform shapes differed.

The noise measurement showed the main frequency components, rounding waveform. The 10kHz setting allowed more harmonics though and preserved the sharper edges. The outputs were 1.1455 V Peak to Peak at 50 Hz and 1 V at 10 kHz, but this difference alone does not prove a gain error because the waveform was different.

The noise measurement showed a main frequency component of 60.98 Hz, which suggests interference from the 60 Hz electrical supply. At ×5000 gain, the output noise measured 206 mV peak-to-peak. This shows why noise matters when recording small physiological

signals: high gain can make unwanted interference significant alongside the desired signal. Since only one noise condition was reported, more measurements would be needed to compare the effects of different gains and filter settings.

6. Conclusion:

This experiment showed how gain, filtering, clipping, and noise affect the signals recorded from the ETH-256 amplifier. The measured ×10 and ×100 gains were about 7% below their nominal values, while the ×1000 result needs to be checked for possible scaling or measurement errors. Most measured filter cutoffs were close to their selected settings, including a cutoff of 50.6 Hz for the nominal 50 Hz filter. The frequency sweep showed increasing attenuation above this cutoff, and the square-wave test showed how removing higher harmonics rounds the output waveform.

The noise measurement identified a 60.98 Hz component and 206 mV peak-to-peak noise at ×5000 gain, suggesting that mains interference affected the recording. Clipping was also observed, but the available measurements were not enough to determine the saturation limits. The experiment demonstrated why amplifier labels alone are not enough to guarantee accurate measurements. Checking the input and output signals, choosing suitable acquisition settings, and accounting for noise are necessary for reliable biomedical recordings.